Lab.Local Enterprise AI & Web Portal Architecture

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Executive Summary

High level system description

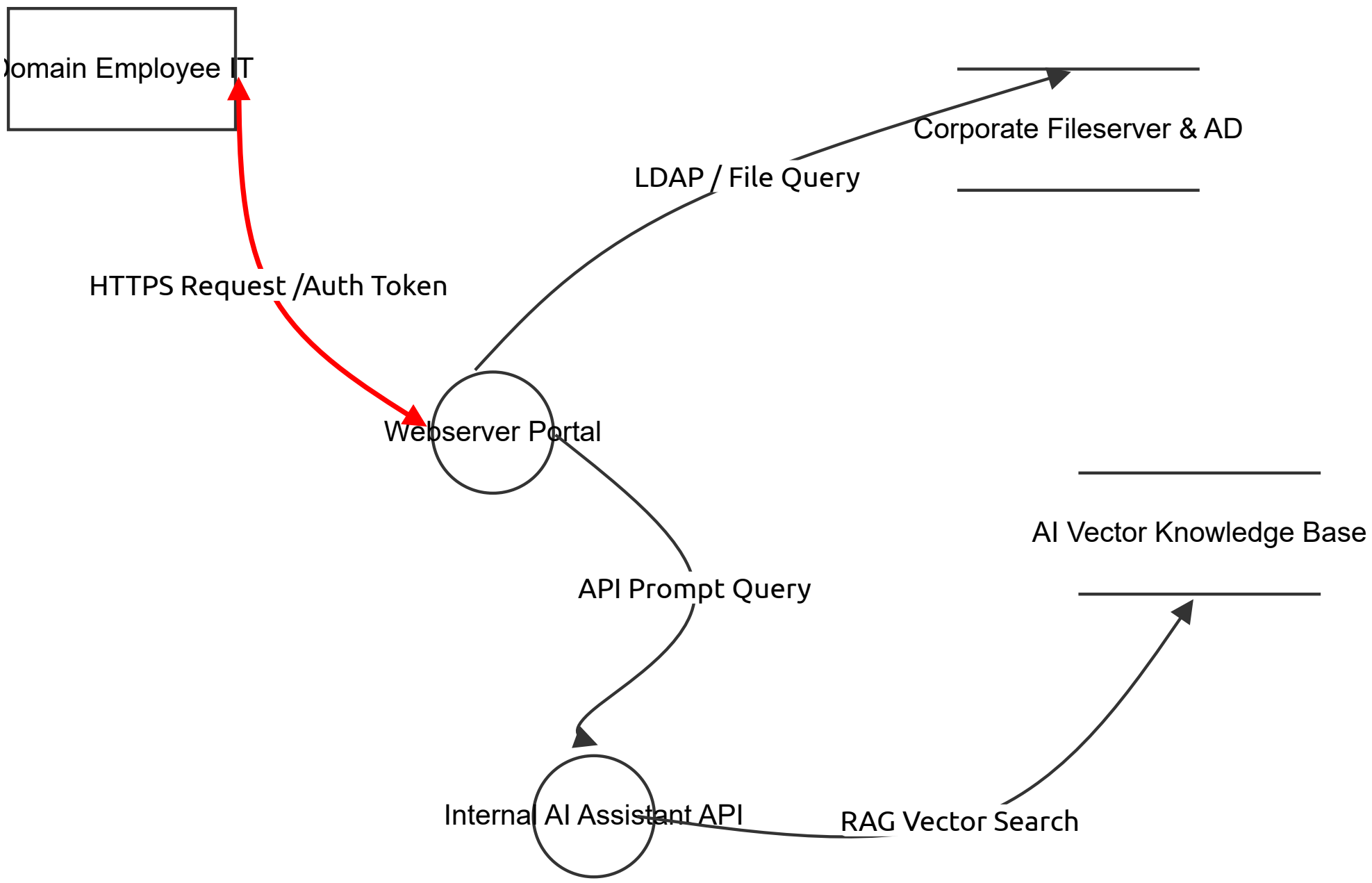
STRIDE and OWASP LLM threat model for internal web application and AI RAG assistant across lab.local VLANs

Summary

Total Threats	4
Total Mitigated	3
Total Open	1
Open / Critical Severity	0
Open / High Severity	1
Open / Medium Severity	0
Open / Low Severity	0

Internal AI & Web Data Flow Diagram

STRIDE and OWASP LLM threat model for internal web application and AI RAG assistant across lab.local VLANs



Internal AI & Web Data Flow Diagram

Domain Employee IT (Actor)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Webserver Portal (Process)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
1	Credential Suffing & Session Spoofing	Spoofing	High	Mitigated		Attacker attempts to brute-force domain credentials or spoof session tokens on the web portal.	Enforced 5-attempt AD Account Lockout GPO on DC (192.168.2.10) and Event ID 4740 alerting in Splunk/Wazuh.

Internal AI Assistant API (Process)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
3	Prompt Injection & Model Abuse (OWASP LLM01)	Elevation of privilege	High	Mitigated		Malicious user crafts input prompts to bypass guardrails and extract sensitive HR/Finance records.	Implemented strict RBAC validation, input sanitization, and network-level DNS sinkholing (0.0.0.0) for external AI endpoints.

Corporate Fileserver & AD (Store)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
4	Unauthorized Data Exfiltration & PII Disclosure	Information disclosure	High	Mitigated		Sensitive company data stored in the fileserver could be accessed or leaked across untrusted VLANs.	Segmented network traffic via pfSense firewall rules and enabled Wazuh FIM (File Integrity Monitoring) / Splunk log aggregation.

AI Vector Knowledge Base (Store)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

HTTPS Request /Auth Token (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
5	Man-in-the-Middle	Tampering	High	Open		Unencrypted	Enforce mutal TLS

API Prompt Query (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

LDAP / File Query (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

RAG Vector Search (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations